Random Matrices

M1R Project

Dr Sheehan Olver

What are random matrices?

Random matrices and their eigenvalues describe various phenomena

Universality: Many families of random matrices have the same eigenvalue statistics.

Free Probability: Adding or multiplying random matrices are described by a version of convolution associated with non-commuting random variables.

Randomised Linear Algebra: Randomness can be used to reduce linear algebra problems to lower dimensional versions which are faster to solve.

Who am I?

Dr Sheehan Olver

- PhD in Cambridge followed by
  Junior Research Fellow at St. John’s College, Oxford
- Researcher in numerical analysis / scientific computing studying:
    - Computational complex analysis
    - Random matrix theory
    - Partial/fractional differential equations
- Won the Adam’s Prize in 2012 for developing numerical methods for Riemann–Hilbert problems, which describe random matrix statistics

Universality

Wigner Ensembles are symmetric random matrices of the form

$$A = B + B^\top$$

where B has independent identically distributed (iid) entries.

Gaussian Orthogonal Ensemble (GOE) is the special case where the entries of B are normally distributed.

What can we say about the statistics of eigenvalues of A?

Universality is the phenomena that certain eigenvalue statistics are, under reasonable constraints, independent of the distribution of B. This includes both global and local statistics.

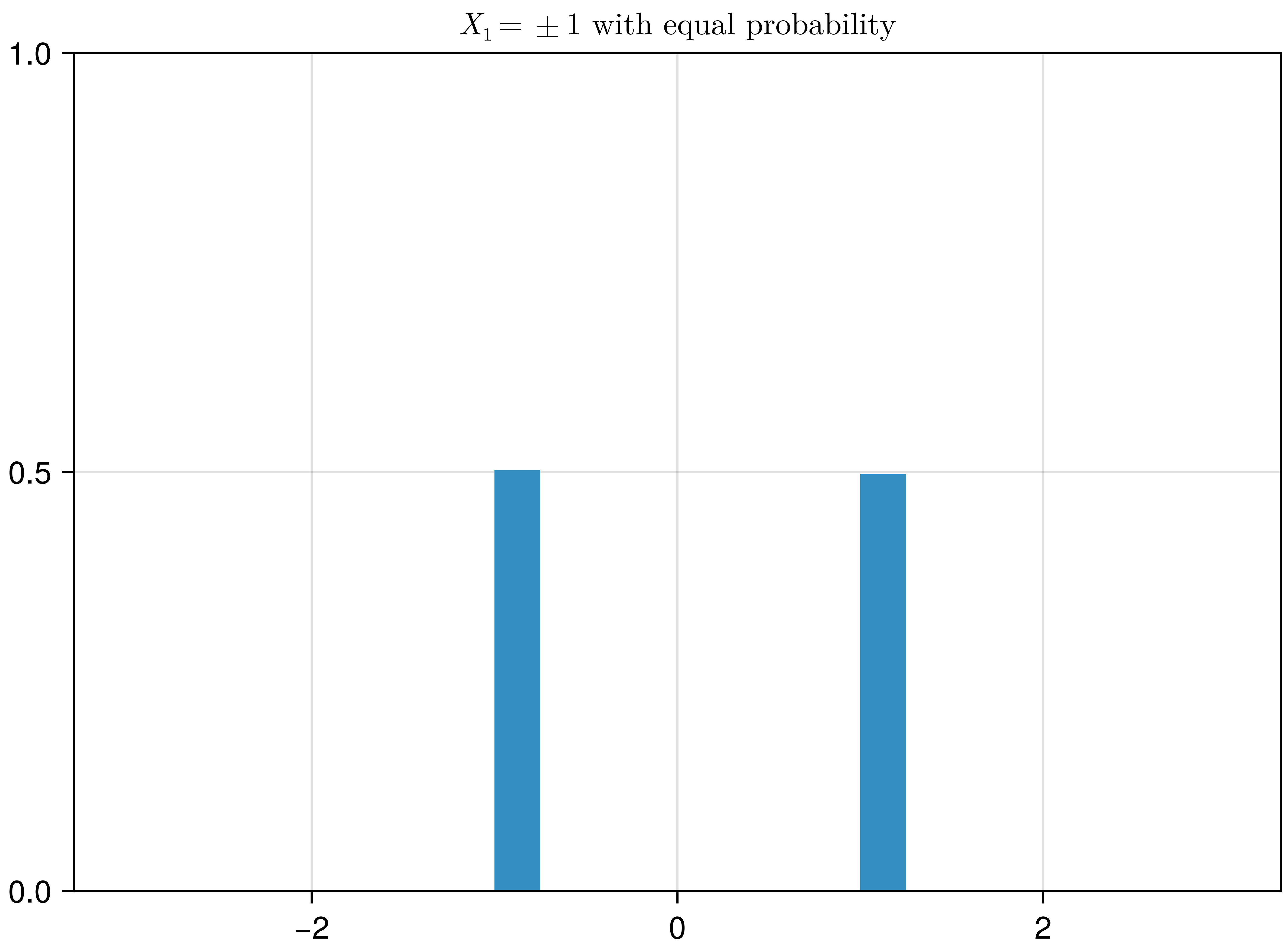
Central Limit Theorem (CLT)

Adding random variables becomes normal

Consider adding and rescaling n samples of iid random variables X_k with zero mean:

$$\frac{X_1 + \cdots + X_n}{\sqrt{n}}$$

If the distribution of X_k is "nice" then as $n \to \infty$ it will look normal



Global Statistics

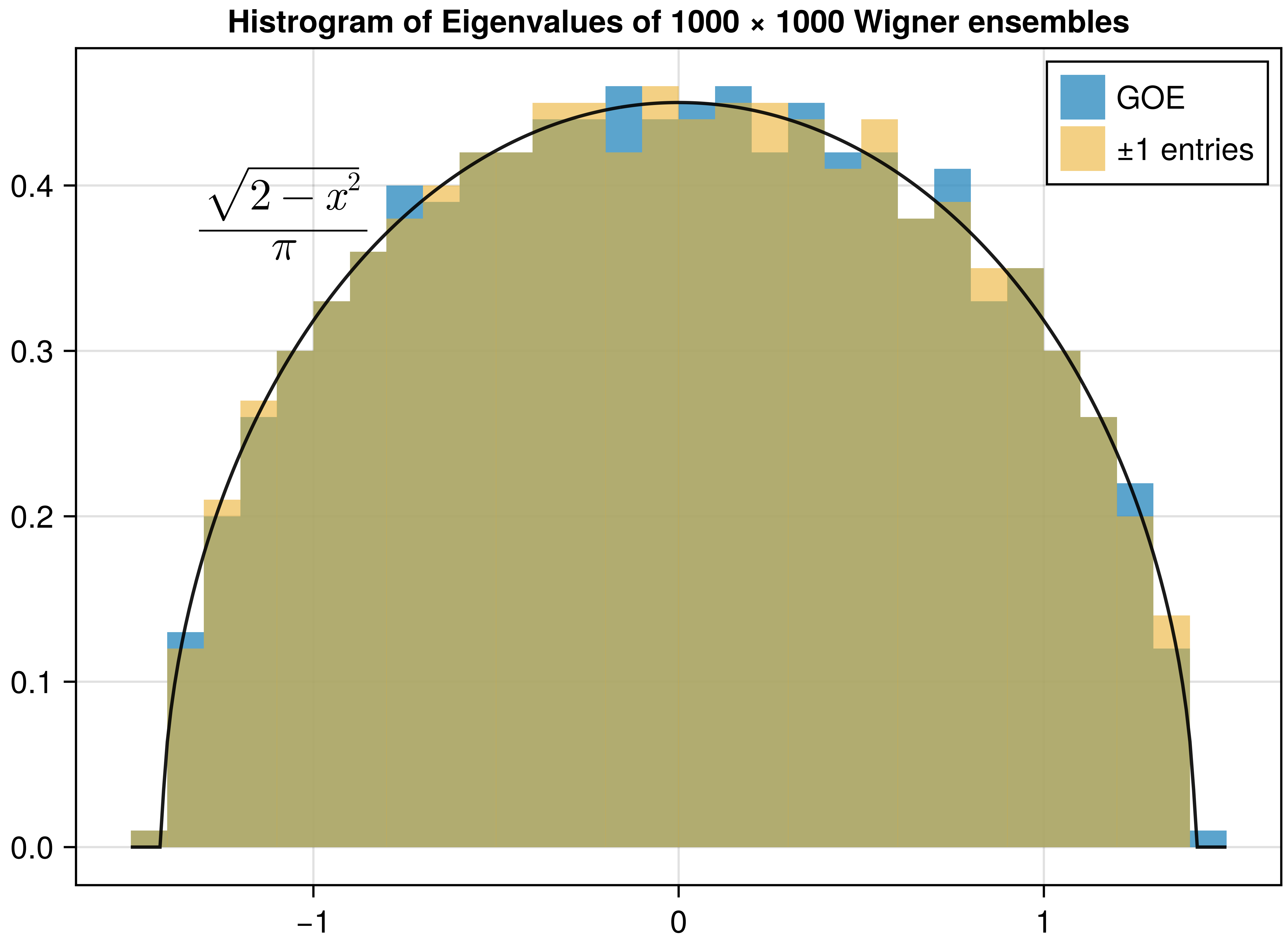
Eigenvalue distributions are universal

Global distribution of eigenvalues of Wigner ensembles is universal:

A semicircle provided the first moments of entries are finite.

Goes back to Wigner in the 1950s.

More recent work by Erdős–Schlein–Yau.



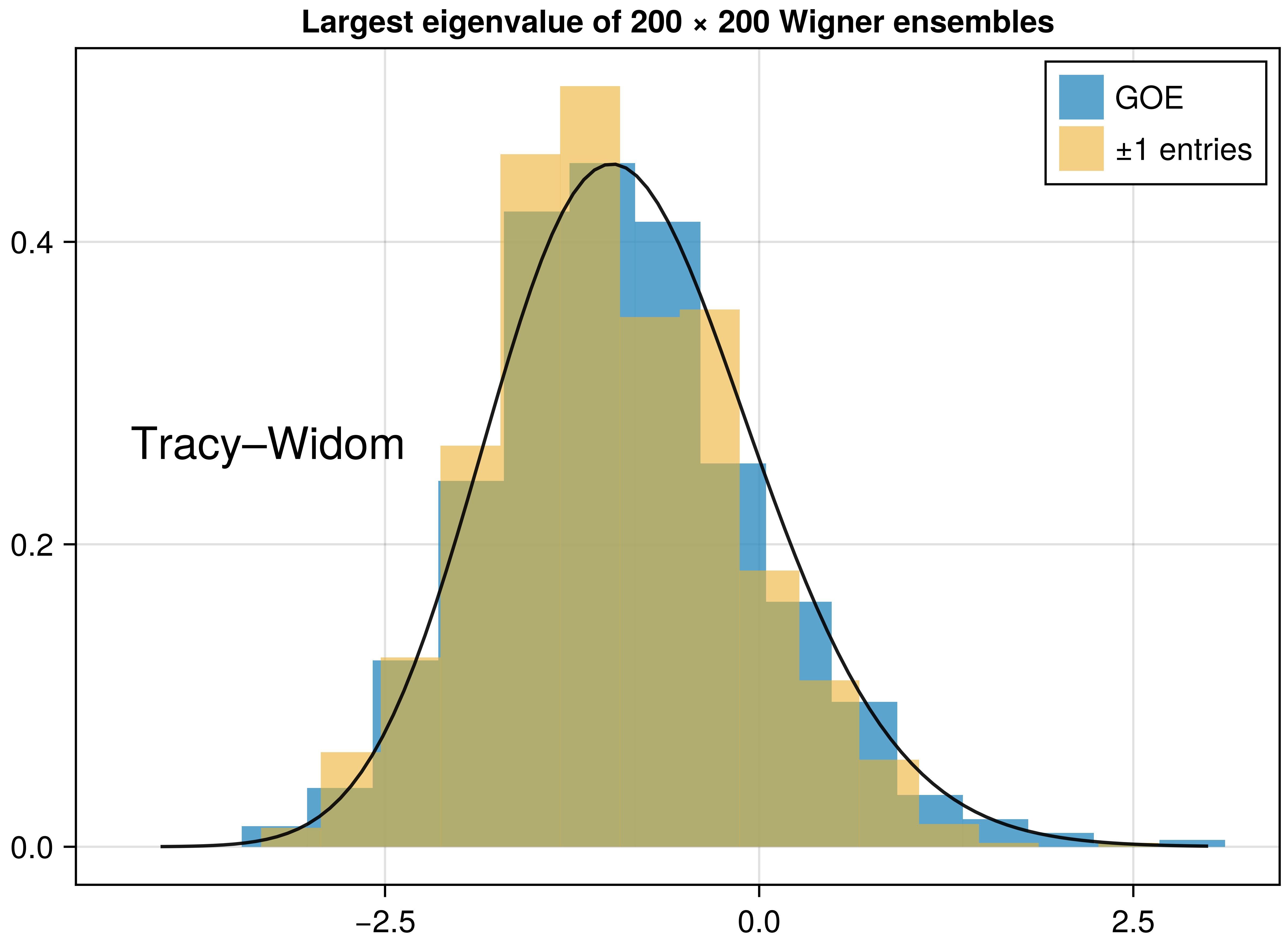
Local Statistics

Eigenvalue spacings are universal

Spacing of eigenvalues or distribution of largest eigenvalue of Wigner ensembles is universal.

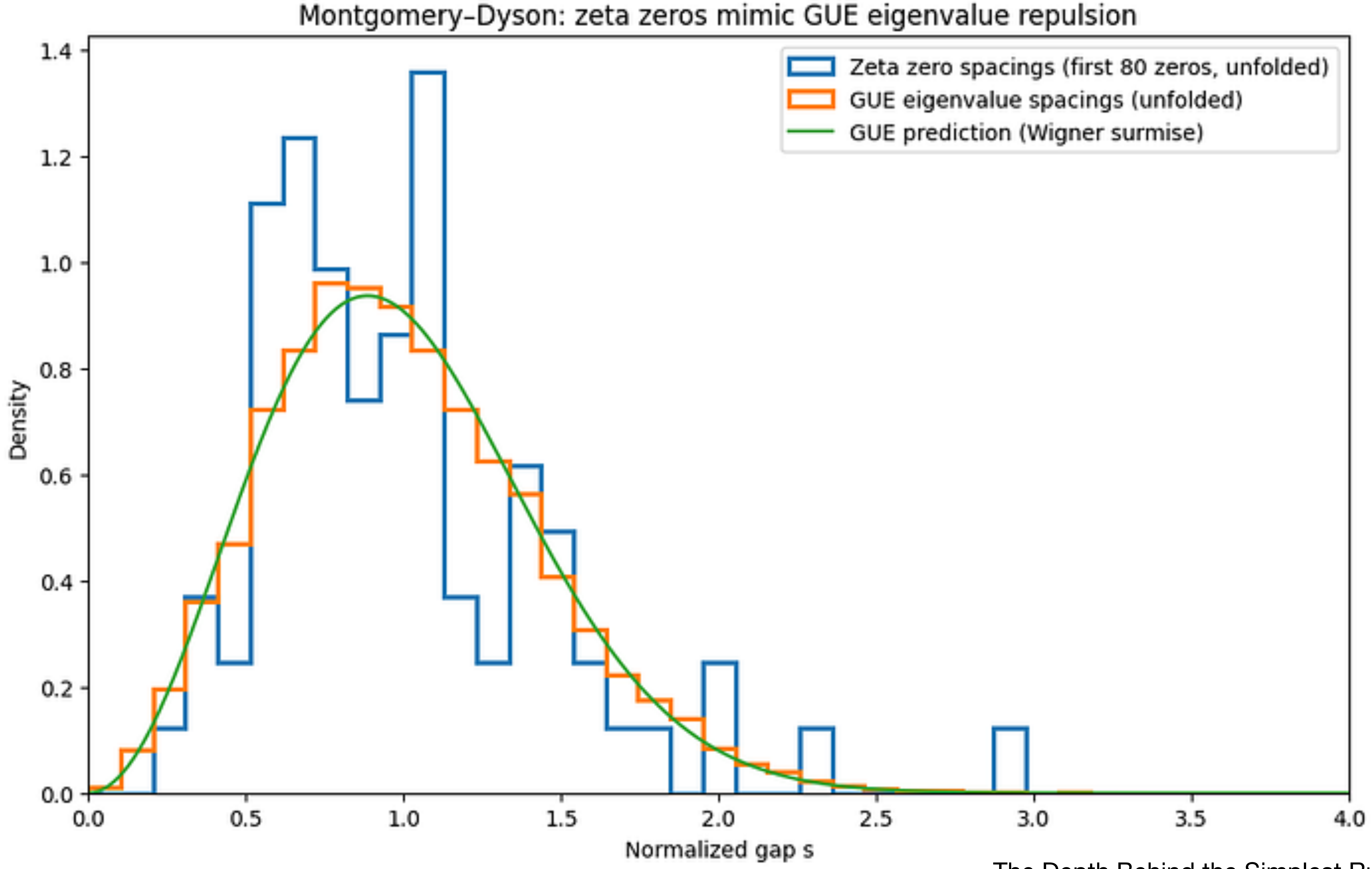
Proved by Tao and Vu (2008–11).

Also describes broad phenomena where there is repulsion.



Connection with Number Theory

Spacing of roots of the Riemann Zeta Function follow universality



The Depth Behind the Simplest Rules: A Chronicle of Prime Numbers | by Daniel Rodríguez | The Quantastic Journal | Medium

Universality poster ideas

Some examples of projects

Can you confirm the connection with zeros of Riemann Zeta function to higher accuracy?

What happens when the 4-moment condition breaks down? I.e. what are eigenvalue statistics for matrices with heavytailed entries?

How connected does a random network need to be to exhibit universality?

Does universality arise in applications where there is repulsion, eg in quantum physics?

Do weight matrices in Neural Networks behave like random matrices? Is there a transition to non-universality behaviour when the training converges?

Free Probability

Algebra with random matrices

Free probability tells eigenvalue distribution of addition/multiplication

Consider the eigenvalues of the sum of two “nice” random matrices

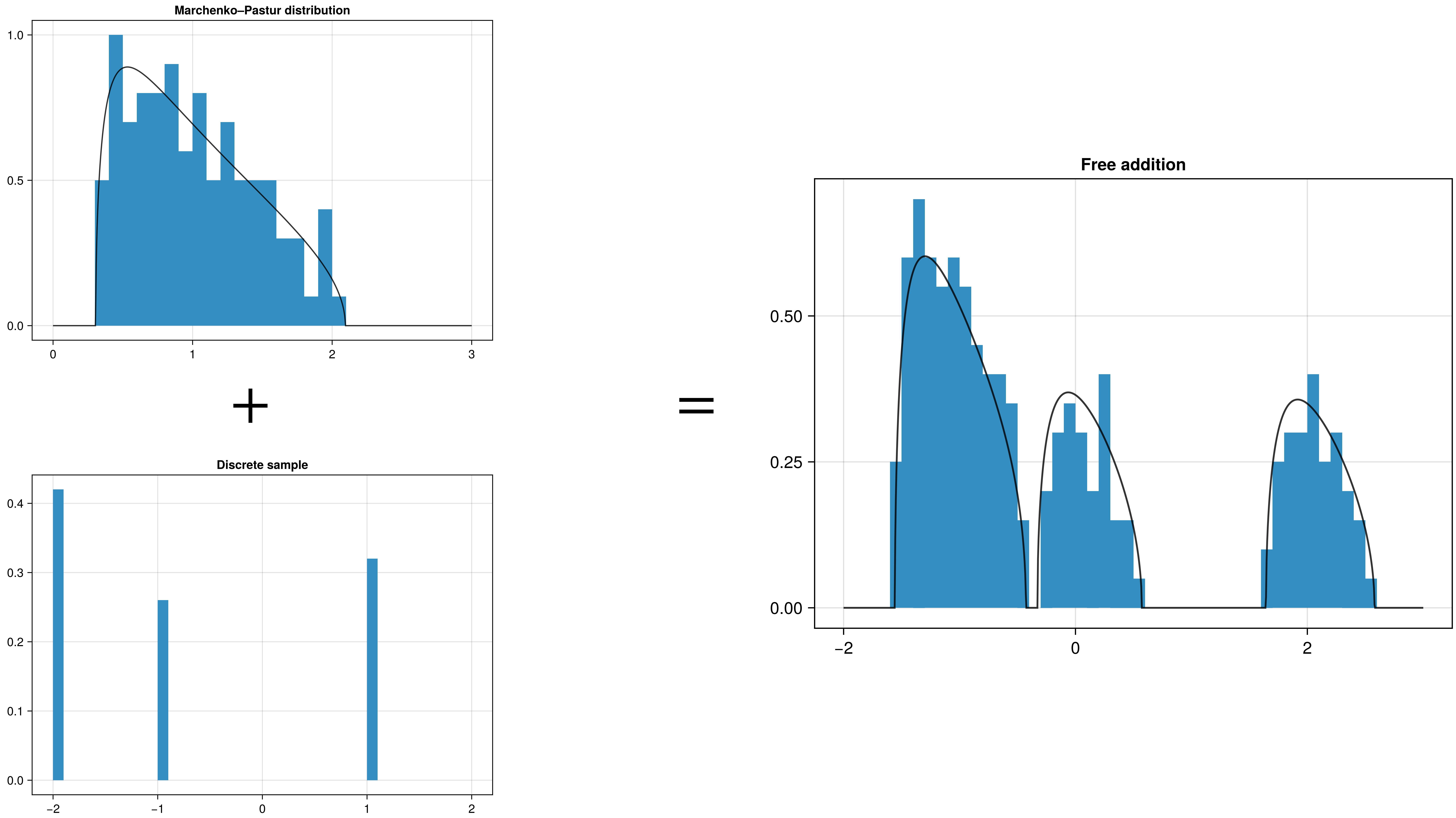
$$A + B$$

The moments of the eigenvalues λ_j of $A + B$ are given by

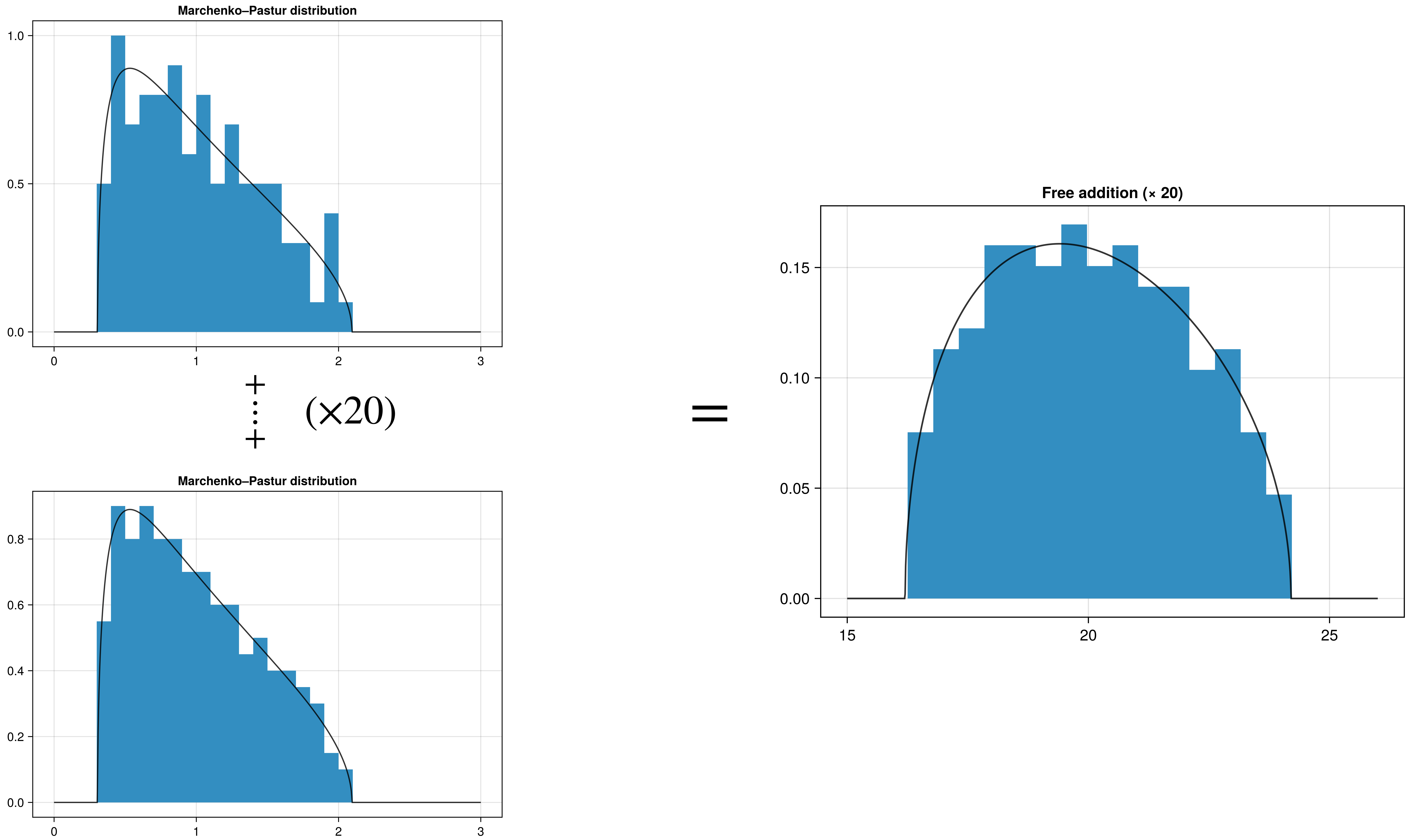
$$\mathbb{E}[\lambda_1^k + \cdots + \lambda_n^k] = \mathbb{E}[\mathrm{Tr}(A + B)^k]$$

Free probability tells us that we can express this as a function of the moments of the eigenvalues of A and B:

$\mathbb{E}[\mathrm{Tr}\ A^k]$ and $\mathbb{E}[\mathrm{Tr}\ B^k]$



Marchenko–Pastur distribution
1.0
0.5
0.0
0
1
2
3
+
Discrete sample
0.4
0.3
0.2
0.1
0.0
−2
−1
0
1
2
=
Free addition
0.50
0.25
0.00
−2
0
2



Marchenko–Pastur distribution
1.0
0.5
0.0
0
1
2
3
+
+
(×20)
Marchenko–Pastur distribution
0.8
0.6
0.4
0.2
0.0
0
1
2
3
=
Free addition (× 20)
0.15
0.10
0.05
0.00
15
20
25

Free probability poster ideas

Some examples of projects

Understanding all free probability operations including multiplication and truncation.

What are the heavytail analogues of the Semicircle Law?

Do additions and multiplications of random matrices still exhibit universality in local statistics?

What are the applications in wireless communication and elsewhere?

A Neural Network can be viewed as a sequence of matrix-vector products, with activator functions in the middle. Can these be connected to free probability?

Randomised Linear Algebra

Introducing randomness for linear algebra problems

Can solve problems with high probability using much smaller matrices

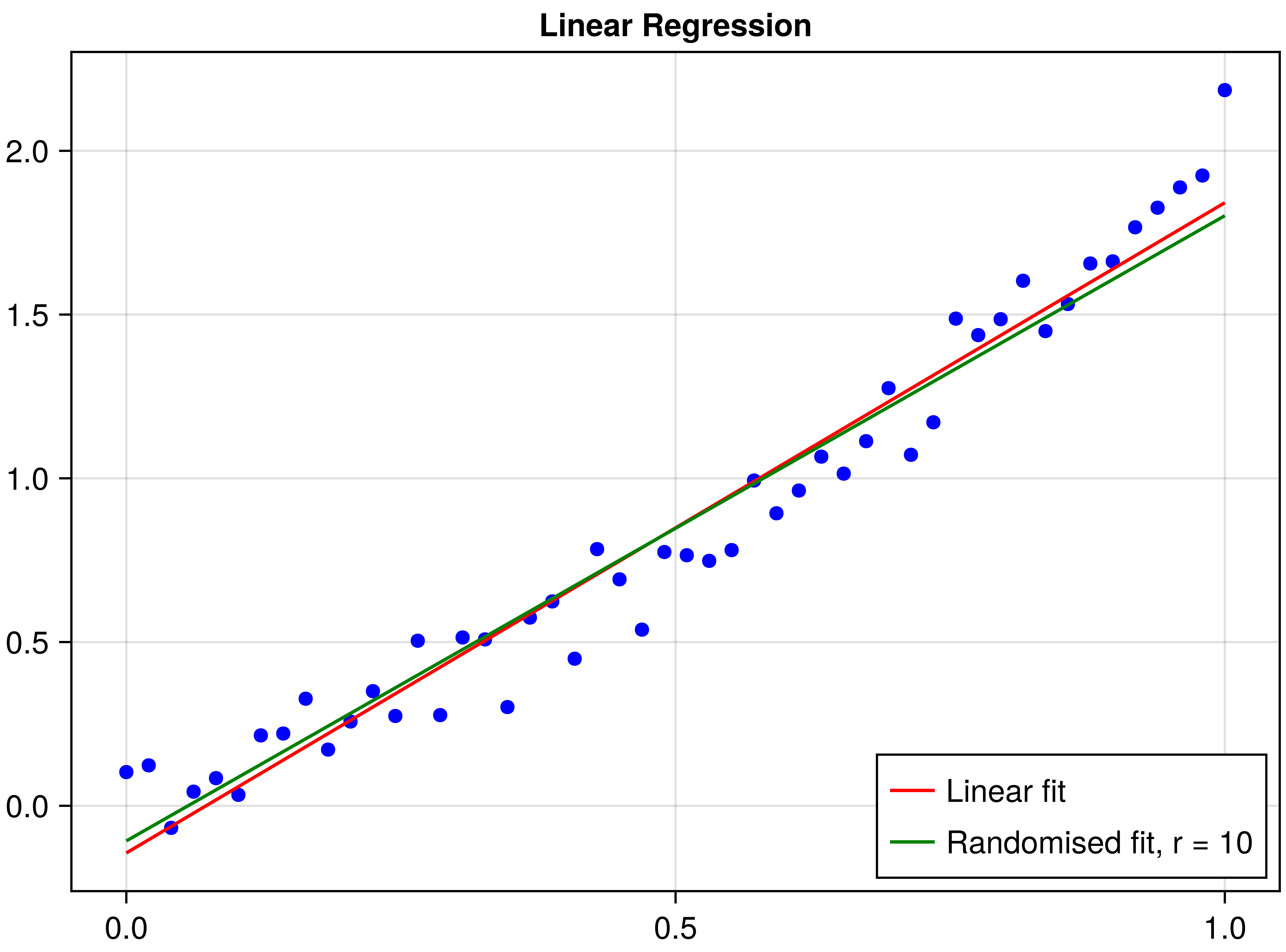
Probabilistic algorithms are becoming increasingly used for high-dimensional problems.

A canonical example is least squares: given a rectangular matrix $A \in \mathbb{R}^{m \times n}$ and right-hand side $\mathbf{b} \in \mathbb{R}^m$, find $\mathbf{x} \in \mathbb{R}^n$ so that $A\mathbf{x} \approx \mathbf{b}$ by minimising:

$$\|A\mathbf{x} - \mathbf{b}\|.$$

Can we get good accuracy using random sketch matrix $P \in \mathbb{R}^{r \times m}$ with $r \ll m$ and minimising

$$\|PA\mathbf{x} - P\mathbf{b}\|?$$


Linear Regression
2.0
1.5
1.0
0.5
0.0
0.0
0.5
1.0
Linear fit
Randomised fit, r = 10

Randomised Linear Algebra poster ideas

Some examples of projects

How does the approximation property depend on the distribution of the sketch matrix?

What other algorithms are amenable to randomisation (low rank approximation, norm estimation, multiplication)?

Can averaging over multiple sketch matrices be used to improve accuracy while preserving speed advantage?

What applications are there to randomised linear algebra such as regression?

Conclusions

Random matrices exhibit a number of interesting phenomena in their eigenvalues and applications in numerical linear algebra.

Universality explores the statistics of random matrices and how they are largely independent of the entries.

Free probability describes the behaviour of addition and multiplication of random matrices.

Randomised linear algebra is an exciting new area using random matrices to achieve faster algorithms.

Each topic has a number of possible poster projects, either investigating the mathematics, exploring experimentally via Monte Carlo simulations, or digging deeper into the applications.